Adding and Subtracting in Scientific Notation

Question Bank 4.01

Name_____
Date_____
Period_____

Show your work. Circle your final answer.

For questions 1 through 6, add or subtract. Write each answer in scientific notation.

1. $(2.4 \times 10^5) + (3.1 \times 10^5)$

2. $(8.3 \times 10^{-4}) - (2.1 \times 10^{-4})$

3. $(6.7 \times 10^9) + (5.8 \times 10^9)$

4. $(9.1 \times 10^4) - (8.6 \times 10^4)$

5. $(7.2 \times 10^6) + (4.3 \times 10^5)$

6. $(5.5 \times 10^{-3}) - (8.0 \times 10^{-5})$

Real quantities.

7. Neptune's diameter is about 4.95×10^{4} kilometers. Saturn's is about 1.21×10^{5} kilometers.

a. What is the sum of the two diameters?

b. Jupiter's diameter is about 1.43×10^{5} kilometers. Is your answer to part (a) greater than Jupiter's diameter?

8. A city's population is about 4.62×10^{5} . The surrounding county's population is about 1.38×10^{6} .

a. What is the total population of the city and the county?

b. How many more people live in the county than in the city?

Working backwards.

9. Find the missing number in each equation.

a. $(3.6 \times 10^7) + (n \times 10^7) = 1.01 \times 10^8$

b. $(8.4 \times 10^{-6}) - (n \times 10^{-6}) = 2.9 \times 10^{-6}$

c. $(2.5 \times 10^4) + (7.5 \times 10^4) =$ _____

Reading someone else's work.

10. Priya writes $(2.6 \times 10^4) + (5.1 \times 10^6) = 7.7 \times 10^{10}$. Name the two mistakes she made, and give the correct sum.

11. Which is the value of $(4.7 \times 10^{-5}) + (8.6 \times 10^{-5})$, written in scientific notation?

- ○ A. 13.3×10^{-10}
- ○ B. 1.33×10^{-4}
- ○ C. 1.33×10^{-10}
- ○ D. 13.3×10^{-5}

12. Select ALL of the expressions whose value is 1.2×10^5 .

- ☐ A. $(1.25 \times 10^5) - (5 \times 10^3)$
- ☐ B. $(1.3 \times 10^5) - (1 \times 10^5)$
- ☐ C. $(1.3 \times 10^5) - (1 \times 10^4)$
- ☐ D. $(8 \times 10^4) + (4 \times 10^5)$
- ☐ E. $(8 \times 10^4) + (4 \times 10^4)$
- ☐ F. $(1.1 \times 10^5) + (1 \times 10^4)$